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EMERGENT MANIFOLD SEPARABILITY DURING REASONING IN LARGE LANGUAGE MODELS

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ABSTRACT

Chain-of-Thought (CoT) prompting significantly improves reasoning in Large Language Models, yet the temporal dynamics of the underlying representation geometry remain poorly understood. We investigate these dynamics by applying Manifold Capacity Theory (MCT) to a compositional Boolean logic task, allowing us to quantify the linear separability of latent representations without the confounding factors of probe training. Our analysis reveals that reasoning manifests as a transient geometric pulse, where concept manifolds are untangled into linearly separable subspaces immediately prior to computation and rapidly compressed thereafter. This behavior diverges from standard linear probe accuracy, which remains high long after computation, suggesting a fundamental distinction between information that is merely retrievable and information that is geometrically prepared for processing. We interpret this phenomenon as *Dynamic Manifold Management*, a mechanism where the model dynamically modulates representational capacity to optimize the bandwidth of the residual stream throughout the reasoning chain.

1 Introduction

Chain-of-Thought (CoT) prompting is highly effective at enabling Language Models (LMs) to solve complex compositional problems [Wei et al., 2022]. The intuition is that CoT distributes the computational load: rather than relying solely on feed-forward depth, the model utilizes the recursive generation of tokens to process information over time. However, the precise mechanisms governing this temporal processing remain largely uncovered.

In studying the information flow in feed-forward computation, investigating how the representation geometry evolves over layers has been proven to provide critical insights [Cohen et al., 2020, Pope et al., 2021]. However, not much is currently known about the evolution of the representation geometry over tokens. We bridge this gap by borrowing geometrical measures from computational neuroscience and deep neural networks (DNN) to track how internal computation unfolds over time.

Existing research on CoT internal representations often focuses on whether the textual output faithfully reflects the model's hidden computation [Turpin et al., 2023, Lanham et al., 2023]. Recent works have used linear probes to show that intermediate answers are often linearly decodable well before they are generated [Zhang et al., 2025, Afzal et al., 2025]. However, standard probing techniques have inherent limitations that may lead to overestimating a model's actual knowledge Afzal et al. [2025], and even show high accuracy when training on random features [Zhang and Bowman, 2018, Wieting and Kiela, 2019].

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To address these limitations, we analyze the latent space using Manifold Capacity Theory (MCT) [Chung et al., 2018] alongside linear probing. Unlike probing, MCT quantifies linear separability as an intrinsic geometric property, eliminating the need to train a classifier. This framework has provided key insights into information processing in visual and speech systems and early language models [Chou et al., 2024, Stephenson et al., 2019, 2021, Mamou et al., 2020, Kirsanov et al., 2025]

In this work, we use MCT to map the dynamics of representation manifolds in the residual stream during a compositional reasoning task (Figure 1).

Our core contributions are:

1. Geometric evidence, provided by Capacity Theory, that CoT involves active reasoning throughout generation, rather than only mimicking a reasoning trace.
2. LM dynamically modulates the representation geometry such that only the concepts that are relevant for a computation at a given time have high manifold capacity. We find that this mechanism keeps the representation dimensionality low, saving the bandwidth budget of the residual stream.

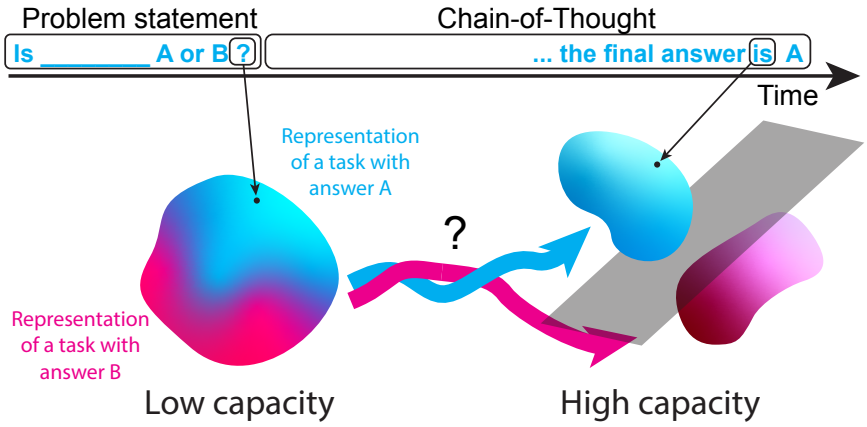

Figure 1: **Manifold Untangling during Chain-of-Thought.** Before any token is generated (left) the latent representations of tasks corresponding to answers A and B are entangled and hard to distinguish resulting in low capacity. As the model generates the CoT, it progressively untangles these representations. By the final token (right), the two manifolds are linearly separable and capacity is high.

2 Related Work

2.1 Internal Mechanisms of Chain-of-Thought

While the performance benefits of CoT are well-established, the nature of the underlying internal computation remains debated. One line of inquiry questions the *faithfulness* of the generated reasoning, suggesting that CoT may be a post-hoc rationalization rather than the actual causal driver of the prediction [Turpin et al., 2023, Lanham et al., 2023]. Smaller, distilled models may be particularly at risk of that, and it has been shown that for small models, more CoT granularity does not necessarily improve performance [Chen et al., 2025]. Conversely, recent interpretability work has identified a circuit within a two-layer attention-only model that plays a role in inductive reasoning [Elhage et al., 2021, Olsson et al., 2022].

A growing body of recent literature focuses specifically on decoding the reasoning states directly from the latent activations. Zhang et al. [2025] and Afzal et al. [2025] investigate the necessity of a long CoT and demonstrate that the correctness of an intermediate step is often linearly decodable long before the token is generated, suggesting that the model maintains a coherent internal truth state, and resulting in proposed early-stopping strategies.

Our work contributes to this discourse by looking for global geometric signatures of the reasoning steps, and specifically investigates the dynamics of the geometric representation along the CoT.

2.2 Linear Representations and Probing Challenges

The prevailing view in mechanistic interpretability is the *Linear Representation Hypothesis*, which suggests that neural networks encode semantic concepts such as sentiment, syntax, or factual truth—as vectors in a

linear subspace [Elhage et al., 2022]. Under this hypothesis, linear probes, supervised classifiers trained on frozen activations, have become a popular tool. They have been used to decode various concepts, from syntactic tree depth [Hewitt and Manning, 2019] to the truthfulness of model statements [Azaria and Mitchell, 2023, Burns et al., 2024, Marks and Tegmark, 2024], and even spatiotemporal coordinates [Gurnee and Tegmark, 2024], serving as a powerful diagnostic tool for locating information within the residual stream.

However, despite their empirical success, relying solely on probing accuracy presents significant methodological challenges. First, the probe is sensitive to regularization, optimizer, early stopping, class imbalance handling, and train/test splits. Moreover, Zhang and Bowman [2018], Wieting and Kiela [2019], and Hewitt and Liang [2019] established that linear probes can achieve high accuracy on random control tasks by exploiting the high-dimensional nature of the representation space or the structural priors of the model architecture. This has prompted the need to find a more geometric approach to interpretability research Afzal et al. [2025].

2.3 Manifold Theory and Representational Geometry

Geometry-based analysis techniques have been actively developed and widely used for studying the brain and the feed-forward networks [Pope et al., 2021, Lehky et al., 2014, Willett et al., 2021]. This approach treats a set of neural representations of inputs as a single entity: a neural manifold. The perspective that the goal of neural computation is to disentangle these manifolds (e.g., think of manifolds of dog images and cat images being disentangled in a vision network for the linear readout) has provided critical insights into the internal mechanism of the brain [DiCarlo and Cox, 2007].

This geometric perspective has been formalized through Manifold Capacity Theory [Chung et al., 2018, Cohen et al., 2020, Chou et al., 2024]. Manifold capacity quantifies the linear separability of these structures by applying principles from statistical mechanics. It is defined as the critical ratio $\alpha = P/N^*$, where P is the number of manifolds and N^* is the minimum dimensionality (of a random subspace to which the manifold is projected) required to achieve linear separation with a 50% probability. The theory also shows that α is dependent on task-dependent geometric quantities, such as dimensionality and radius, providing a direct insight into what geometric properties of manifolds determine an observed α. Essentially, it measures the efficiency of task-relevant feature encoding from the perspective of a linear downstream decoder. Crucially, by treating separability as an inherent geometric property, this framework eliminates the methodological pitfalls of probing.

This framework has previously been shown to provide useful insights into feedforward computation in the visual and auditory cortex [Chou et al., 2024], as well as in artificial vision models, where class manifolds are observed to progressively *untangle* over layers [Stephenson et al., 2019]. Figure 1 shows a schematic depiction of such untangling. However, its application to Large Language Models remains nascent. While Mamou et al. [2020] analyzed the geometry of static word embeddings and recent work has begun to explore the intrinsic dimension of transformer representations [Kirsanov et al., 2025], we extend this lineage by applying Manifold Capacity to the temporal dimension, tracking the dynamic modulation of concept manifolds during reasoning.

3 Methods

3.1 Compositional Boolean Logic Task

As a representative benchmark for tasks requiring multi-step reasoning, we utilize a hierarchical Boolean logic task. The goal is to evaluate the Boolean value of a nested expression, such as `((True and False) or (True xor True))`. Each expression corresponds to a full binary tree of height h, where internal nodes represent logical operators `AND, OR, XOR` and leaf nodes provide Boolean constants.

To facilitate a structured analysis of the internal representations, we employ a consistent naming and ordering scheme. Each internal node is assigned a unique identifier `[ID]` based on a post-order or level-order traversal of the logic tree. This ensures that, with an appropriate prompt strategy, all tasks are solved in the same order. For our primary analysis on the internal representation, we generate a balanced dataset of 256 expressions (128 True, 128 False at the root) for a tree of height $h = 5$, containing $2^h - 1 = 31$ internal nodes.

3.2 Prompting Strategy and Alignment

To capture the evolution of neural manifolds, we require the model to adhere to a rigid, structured output format using Markdown structural tokens. We prompt the model to reason through three distinct phases: *Problem Statement* (restating the expression and node IDs), *Solve* (step-by-step evaluation), and *Summary* (recalling all intermediate results).